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NORME DE LA CEI

INTERNATIONAL ELECTROTECHNICAL COMMISSION
IEC STANDARD

Modification n° 1

Mai 1984
à la

Amendment No. 1

May 1984
to

Publication 516
1975

**Système modulaire d'instrumentation pour le traitement de l'information ;
système CAMAC**

**A modular instrumentation system for data handling ;
CAMAC system**

Les modifications contenues dans le présent document ont été approuvées suivant la Règle des Six Mois.

Le projet de modifications, discuté par le Comité d'Etudes n° 45, fut diffusé en janvier 1983 pour approbation suivant la Règle des Six Mois, sous forme de document 45(Bureau Central)167.

The amendments contained in this document have been approved under the Six Months' Rule.

The draft amendments, discussed by Technical Committee No. 45, were circulated for approval under the Six Months' Rule in January 1983, as Document 45(Central Office)167.



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Page 3

Add the following text to the contents page:

— under "Clause":

10. Use of supplementary 6 V power (Bus-lines Y1, Y2)

— under "Figures":

11. Some LAM structure options

12. Typical current sharing with supplementary 6 V power

Page 17

4.4 The Dataway

Delete last sentence of last paragraph: "Relatively high voltages ... neutral).".

Page 19

TABLE I

Replace the last entry by the following:

Additional power lines			
+12 V d.c.	+12	1	
-12 V d.c.	-12	1	
Clean earth	E	1	Reference for circuits requiring clean earth
Supplementary -6 V	Y1	1	See Clause 10
Supplementary +6 V	Y2	1	See Clause 10
Reserved	Undesignated	3	

Pages 21 and 23

TABLES II and III

Replace the last entry by the following:

Power bus-lines	{	- 12 V d.c.	- 12	- 24	- 24 V d.c.	}	Power bus-lines
		Reserved (c)*	-	- 6	- 6 V d.c.		
		Reserved (a)*	-	-	Reserved (b)*		
		Supplementary - 6 V**	Y1	E	Clean earth		
		+ 12 V d.c.	+ 12	+ 24	+ 24 V d.c.		
		Supplementary + 6 V**	Y2	+ 6	+ 6 V d.c.		
		0 V (power return)	0	0	0 V (power return)		

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Add below Tables II and III:

- * Reserved (c) was previously assigned to +200 V d.c.,
Reserved (a) to 117 V a.c. line and,
Reserved (b) to 117 V a.c. neutral.
These assignments have been cancelled in order to avoid hazardous voltages on the connectors.
- ** See Clause 10.

Page 25

5.1.1 *Station number (N)**Add to last paragraph:* “and the design of the modules should permit this.”.

Page 37

5.7 *Power lines**Replace the existing text by the following:*

The Dataway shall include lines for all the mandatory and additional power supplies shown in Table I.

The Dataway need not include lines for the supplementary +6 V and –6 V supplies (Y2 and Y1), but its construction should be such that bus-lines and wiring in accordance with Clause 10 can be installed on Y1 and Y2 if required.

Details of the voltage tolerances and permitted loadings of the power lines are given in Clause 8.

Page 45

7.1.2 *Current standards for Dataway signals**Add the following new paragraph:*

There is no restriction on the number of modules that can be addressed simultaneously (see Sub-clause 5.1.1). The signal output from modules to the Dataway Q, R, and X lines should therefore be capable of operating in intrinsic OR mode (see Sub-clause 7.1). Thus, it should be possible for one module to drive a line to the logic “1” state while all other modules are generating logic “0” outputs to the same line.

8. Power line standards

At the end of the second paragraph add: “(See also Clause 10).”.

Replace the third paragraph by the following:

The additional lines are required for special requirements, for example, compatibility with the NIM system. There are heavy current lines for +12 V and –12 V d.c. and a low current line for an independent and isolated clean earth (ground) return E. The 12 V lines are not necessarily powered unless specifically required for use.

Add Clause 10, as follows:

10. Use of supplementary 6 V power (bus-lines Y1, Y2)

In special cases, plug-in units may be used that draw more than 3 A at +6 V or –6 V.

Such plug-in units are considered special, but they shall comply with the current rating for Dataway connector contacts as given in Note 1 of Table X. The total power dissipation in the station or stations occupied by the plug-in unit shall be consistent with Note 3 of Table X.

They shall draw their excess (supplementary) 6 V power, not to exceed 3 A per contact, through the Y1 and Y2 contacts. The total return current in each 0 V contact, including any current from the 24 V and 12 V lines, shall not exceed 3 A per contact.

In order to meet this requirement, the current distribution between the two 0 V contacts may need to be controlled. An example of a suitable circuit is shown in Figure 12.

Plug-in units requiring supplementary 6 V power should only be installed in special crates in which the Y1 and Y2 contacts are bussed and wired to provide the necessary power (see also Sub-clause 5.7).

Plug-in units requiring supplementary 6 V power shall have the supplementary (Y1, Y2) voltage and current requirements clearly and permanently marked on the front panel.

Crates equipped with Y1 and Y2 bus lines and wired for supplementary 6 V power shall be clearly marked to indicate this.

The mandatory front panel marking showing supplementary power requirements should be reinforced by a conspicuous warning on the plug-in unit (for example, on the side panel), indicating that it can only be used in crates equipped with Y1 and Y2 supplementary power).

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Replace Table X by the following:

TABLE X
Power line standards

Nominal voltage on power line in crate	Voltage tolerance at Dataway connectors	Maximum current loads	
		In the plug-in (per unit width) See Notes 1 and 3 (A)	In the crate See Note 2 (A)
Mandatory			
+24 V d.c.	±1.0%	1 A	6 A
+6 V d.c.	±2.5%	2 A*	25 A*
-6 V d.c.	±2.5%	2 A*	25 A*
-24 V d.c.	±1.0%	1 A	6 A
0 V			
Additional (as required)			
+12 V d.c.	±1.0%		
-12 V d.c.	±1.0%		

1) The current carried by each contact of the Dataway connector must not exceed 3 A.

2) The total power dissipation in a crate without forced ventilation must not exceed 200 W.

3) The power dissipation in each station must not exceed 8 W in general or 25 W under special circumstances.

* See Clause 10.

Page 60

In Figure 5, add the following Note 6:

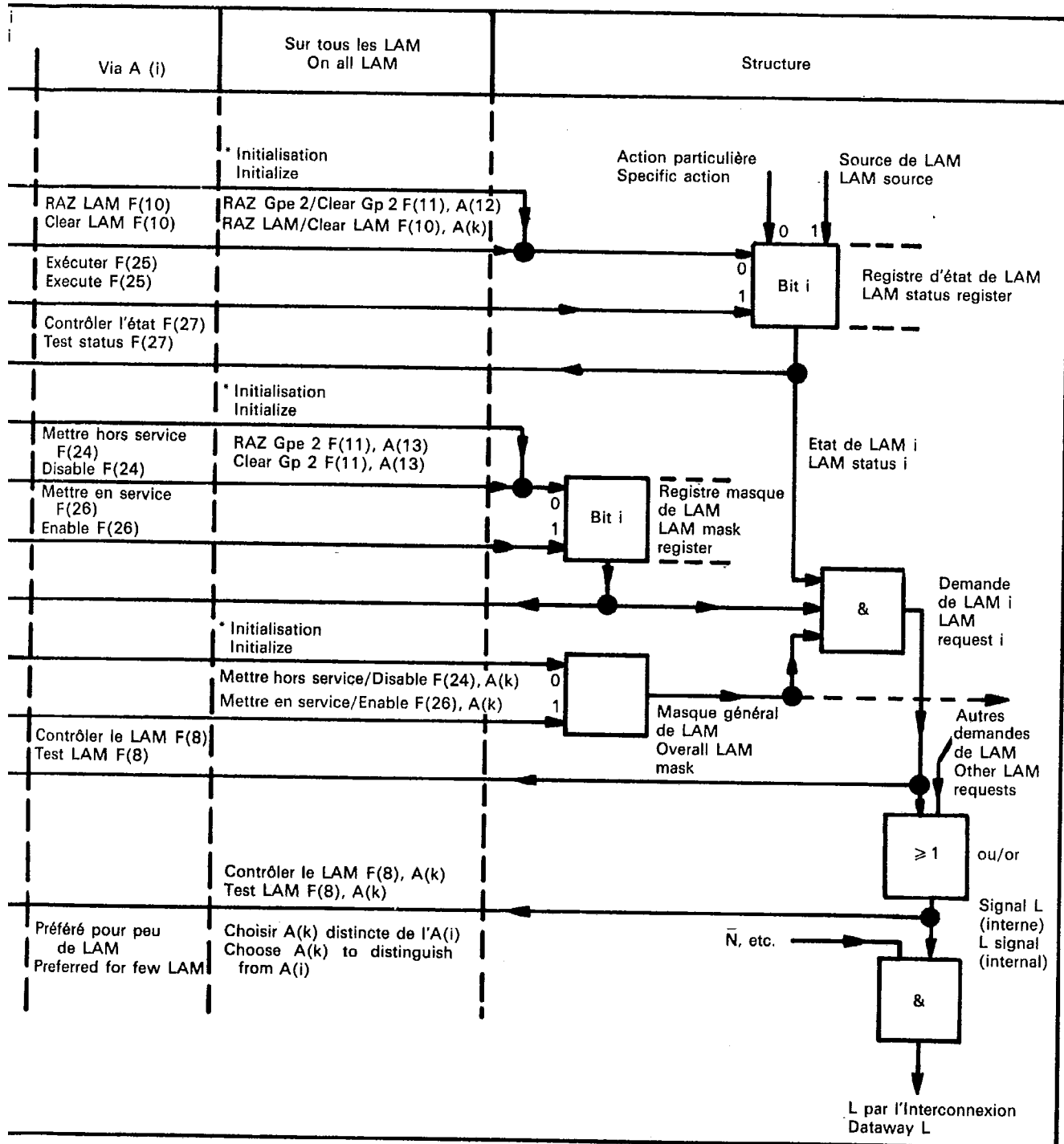
6. — A chamfer of 0.3 mm max. × 0.3 mm max. is permitted on the two vertical edges of the connector even though Section a-a of 5.2 shows zero chamfer.

Pages 64 and 65

In Figures 9 and 10, add the following Notes 5 and 6:

5. — For all signals the minimum rise or fall time is 10 ns. See Sub-clause 7.1.
6. — Signal transition at t_0 or t_2 may be absent if the signals on command or data lines are the same for the immediately preceding or following operations. See Sub-clause 7.1.3.1.

After page 65, add the following new Figures 11 and 12:



033/84

FIG. 11. — Quelques options de structure du circuit d'appel.

* Obligatoire (voir paragraphe 5.4.1.1).

Some LAM structure options.

* Mandatory (see Sub-clause 5.4.1.1).

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Publication 516

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**Système modulaire d'instrumentation pour le traitement de l'information;
système CAMAC**

**A modular instrumentation system for data handling;
CAMAC system**



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INTERNATIONAL ELECTROTECHNICAL COMMISSION

**A MODULAR INSTRUMENTATION SYSTEM
FOR DATA HANDLING;
CAMAC SYSTEM**

FOREWORD

- 1) The formal decisions or agreements of the IEC on technical matters, prepared by Technical Committees on which all the National Committees having a special interest therein are represented, express, as nearly as possible, an international consensus of opinion on the subjects dealt with.
- 2) They have the form of recommendations for international use and they are accepted by the National Committees in that sense.
- 3) In order to promote international unification, the IEC expresses the wish that all National Committees should adopt the text of the IEC recommendation for their national rules in so far as national conditions will permit. Any divergence between the IEC recommendations and the corresponding national rules should, as far as possible, be clearly indicated in the latter.

PREFACE

This publication has been prepared by IEC Technical Committee No. 45, Nuclear Instrumentation.

A first draft was prepared according to document EUR 4100e published in 1972 by ESONE Committee and describing the CAMAC system, a modular system widely used in nuclear laboratories throughout the world for data handling in measurement and control situations.

This draft was discussed at the meeting held in The Hague in 1973, as a result of which the draft, document 45(Central Office)83, was submitted to the National Committees for approval under the Six Months' Rule in February 1974.

The following countries voted explicitly in favour of publication:

Belgium	Poland
Czechoslovakia	Romania
Denmark	South Africa (Republic of)
Finland	Switzerland
France	Turkey
Germany	Union of Soviet
Israel	Socialist Republics
Italy	United Kingdom
Japan	United States of America
Netherlands	Yugoslavia

Other IEC publications quoted in this publication:

- Publications Nos. 297: Dimensions of Panels and Racks (for Nuclear Electronic Instruments).
482: Dimensions of Electronic Instrument Modules (for Nuclear Electronic Instruments).

A MODULAR INSTRUMENTATION SYSTEM FOR DATA HANDLING; CAMAC SYSTEM

1. Object

This publication is intended to define a modular instrumentation system capable of linking transducers and other devices with digital controllers or computers. It consists of mechanical standards and signal standards that are sufficient to ensure compatibility between units from different sources of design and production.

The basic features of CAMAC are summarized as follows:

- a) It is a modular system, with functional units which can be combined to form equipment assemblies.
- b) The functional units are constructed as plug-in units and are mounted in a standard crate.
- c) The mechanical structure is designed to exploit the high component packing density possible with integrated circuit packages and similar devices.
- d) Each plug-in unit makes direct connection to a standard Dataway. This highway forms part of the crate and conveys digital data, control signals and power. The standards of the Dataway are independent of the type of plug-in unit or computer used.
- e) The system has been designed so that an assembly consisting of a crate and plug-in units can be connected to an on-line digital computer. However, the use of a computer is entirely optional and no part of this specification depends upon its presence in the system.
- f) External connections to plug-in units may conform to the digital or analogue signal standards of associated transducers, computers, etc., or to the requirements given in this publication for digital signals.
- g) Several CAMAC crates (up to 7) may be interconnected by the CAMAC branch highway.¹⁾

In order to claim compatibility with the CAMAC specification, any equipment or system must comply with the mandatory rules (see Sub-clause 3.1).

2. Scope

This standard applies to nuclear instrumentation but may be utilized also for other applications. Further IEC standards may indicate such extensions of this scope. For reactor instrumentation and control systems, other packaging of electronic nuclear instruments may also be used.

2.1 This standard applies to systems consisting of modular electronic instrument units that require input/output transfers for the purpose of digital data processing, normally in association with a form of common controller, computer or other automatic data processor.

2.2 This standard applies to word-serial transfers that involve the parallel transmission of up to, and including, 24-bits as a word between one of many sources and one of many acceptors of data.

2.3 This standard does not concern itself with the features, characteristics or requirements of the system inter-connection with the common controller, computer or other automatic data-processor, except in so far as these affect the operating conditions of the interface characteristics of the units.

¹⁾ The CAMAC branch highway is under consideration.

3. Terminology

3.1 Interpretation of this publication

Statements that specify mandatory aspects of the system are written in **bold type** and are usually accompanied by the word “shall”.

The word “should” indicates a recommended or preferred practice which is to be followed unless there are sound reasons to the contrary.

The word “may” indicates a permitted practice, leaving freedom of choice to the designer.

3.2 Definition of “module” and “controller”

In this standard, the terms “module” and “controller” refer to plug-in units whose use, if any, of each Dataway line is consistent with the following table. A controller occupies the control station (see Sub-clause 4.4) and at least one normal station. A module occupies one or more normal stations. A plug-in unit may combine some features of a module with some of a controller.

Line	Use by a module	Use by a controller
A	Receives	Generates
B	Receives	Generates
C	Receives	Generates
F	Receives	Generates
L	Generates	Receives
N	Receives	Generates
Q	Generates	Receives
R	Generates	Receives
S	Receives	Generates
W	Receives	Generates
X	Generates	Receives
Z	Receives	Generates

4. Mechanical characteristics

CAMAC is a modular system. Equipment assemblies are formed by mounting appropriate “plug-in” units in a standard chassis or “crate”. Each plug-in unit occupies one or more mounting “stations” in the crate. At each station there is an 86-way connector socket giving access to the “Dataway”, a data highway which forms part of the crate. The Dataway consists mainly of bus-lines for data, control and power.

Drawings for the manufacture of CAMAC compatible crates and plug-in units may be derived from the definitive dimensions given in Figures 1 to 3, pages 56 to 58, for crates, Figure 4, page 59, for plug-in units, and Figure 5, page 60, for Dataway connector plugs and sockets.

Recommended dimensions for ventilated crates, NIM adaptors and printed wiring cards for plug-in units are given in the non-mandatory Figures 6 to 8, pages 61 to 63, respectively.

All dimensions in these figures are in millimetres unless otherwise indicated.

4.1 The crate

The crate mounts in a standard 19-inch rack and has up to 25 stations for plug-in units on a pitch of 17.2 mm. Each station has upper and lower guides for the runners of a plug-in unit, an 86-way Dataway connector socket, and a tapped hole for the fixing screw of a plug-in unit.

Normal CAMAC plug-in unit described in Sub-clause 4.2 is the type C module of IEC Publication 482, Dimensions of Electronic Instrument Modules (for Nuclear Electronic Instruments). Furthermore Type N module of this IEC Publication (NIM module) can also be mounted in the crate on their basic pitch of 34.4 mm (see Sub-clause 4.3).

Unless otherwise indicated, all crates shall conform to Figures 1 to 3 and those parts of Figure 5 defining the connector socket.

Sub-clauses 4.1.1 and 4.1.2 are comments on these figures.

4.1.1 Dimensions

Figure 1 shows the front view of a basic 25-station crate which occupies the minimum height corresponding to dimension 5U of IEC Publication 297, Dimensions of Panels and Racks (for Nuclear Electronic Instruments).

Crates may have less than 25 stations, which, as indicated by Note 3 on Figure 1, need not be positioned symmetrically.

The lower cross-member of the crate has holes tapped ISO M4 pitch 0.7 for the fixing screws of CAMAC plug-in units, and intermediate holes tapped UNC 6-32 for the lower fixing screws of NIM units. The upper cross-member may also have holes for the upper fixing screws of NIM units. The positions of these holes for CAMAC and NIM units, relative to the left-hand edge of the front aperture, are given in Figure 1 by the formulae for dimensions z and w , respectively.

The positions of the centres of the guides, also relative to the left-hand edge of the aperture, are given by the formula for dimension x in Figure 1. Detail A shows the entry into a guide. The dimensions of the lead-in are not specified.

Detail B gives dimensions and spacing of mounting holes as specified in IEC Publication 297.

Figure 2 is a plan view of the lower guides in the crate. In order to remove any heat generated in the plug-in units, it is necessary to provide adequate ventilation through the bottom and top of the crate. The unobstructed area between adjacent guides, both at the top and bottom of the crate, is not permitted to be less than 15 cm² and should preferably be distributed over the full depth of the crate from the front cross-members to the Dataway assembly. If crates such as that shown in Figure 1 (with height 5U) are mounted above or below other equipment (including other similar crates), it may be necessary to use intermediate deflectors, etc., to ensure adequate ventilation. Alternatively, the crate may be extended to include additional ventilation features, as described in Sub-clause 4.1.3.

Figure 3 is a sectional side view on the offset line $d-d$ in Figure 1, passing through the centre of an upper guide and a ventilating space between lower guides. The front faces of the upper and lower cross-members constitute the vertical datum of the crate. This datum is set back from the front face of the crate by a distance e , typically between 3 mm and 4 mm, so that the front panels of plug-in units do not project beyond the front of the crate. The backs of the crate-mounting flanges are typically, but not necessarily, aligned with the datum.

The front ends of the upper and lower guides may be set back from the vertical datum. The guides extend sufficiently far towards the rear of the crate to ensure that the connector plug of a plug-in unit is guided into the entry of the connector socket.

The minimum overall depth of the crate provides mechanical protection for the Dataway assembly. The side panels are shorter than the frontal height of the crate (see dimensions a in Figures 1, 3 and 6) to permit the use of typical runners for supporting the crate in the rack. This reduction in height extends at least to within 25 mm of the rear face of the rack-mounting flanges of the crate.

The running surface of the lower guide constitutes the crate horizontal datum. The Dataway assembly is not permitted to extend upwards more than 135 mm from this horizontal datum, so that there is unrestricted access to the upper part of the rear of plug-in units.

The positions of the connector sockets are defined with respect to the three datum lines of the crate. The centre lines of the sockets are defined with respect to the left-hand edge of the front aperture by dimension y in Figure 1. The vertical datum of the sockets is shown relative to the vertical datum of the crate in Figures 2 and 3, and the horizontal datum of the sockets relative to the horizontal datum of the crate in Figure 3.

4.1.2 Dataway connector sockets

The Dataway connector sockets have two rows of 43 contacts on a pitch of 2.54 mm (0.1 in). Mandatory and recommended dimensions of the sockets are given in Figure 5, together with additional commonly used dimensions upon which the designs of many existing crates and Dataway assemblies have been based.

The vertical datum of the connector sockets is the nominal position of the leading edge of the connector plug of a plug-in unit fully inserted into the crate. The position of the vertical datum is defined in Figure 5.5 with respect to other functional features of the socket. In some commonly used sockets the plane of the mounting face coincides with the vertical datum of the connector socket, but this is not necessarily so.

The maximum forward projection of the connector socket in front of the vertical datum is shown in Figure 5.5. The shapes of the straight or curved chamfers that guide the connector plug into the socket are shown in Figures 5.6, 5.7 and 5.8. Within the minimum width shown for each chamfer the angle between any tangent to the chamfer and the line of entry of the connector plug shall not exceed 60° .

If the front aperture of the crate extends to the inner surface of the right-hand side panel (as in Figures 1 and 2) the adjacent connector socket cannot exceed the recommended width of 12 mm. Elsewhere, sockets up to the maximum width of 17.2 mm can be used.

The dimensions of the contacts of the connector socket are shown in Figure 5.4. The position of each edge is defined by a dimension (d , D) relative to the horizontal datum of the socket, and is completely independent of the positions of all other edges on both rows of contacts.

Alternatively, a connector socket with point contacts may be used, in which case the distance between each point contact and the horizontal datum of the connector socket is $(2.56 + 2.54 k) \pm 0.13$.

4.1.3 *Optional features of the crate*

The height of the crate may be extended by an integral number of U units ($U = 44.45$ mm), as in Figure 6, in order to provide an entry for cool air, which then flows up between the guides, and an exit for any warm air that may be rising from equipment below.

A crate may have fewer than 25 stations. The width of the front aperture is $17.2 s \pm_{0.0}^{+0.3}$ mm for s stations, and formulae given in Figure 1 are used for locating the guides, connector socket, etc., at each station.

Power supply units may be mounted at the rear of a CAMAC crate. The overall depth of a crate with rear-mounted power supplies may be limited by the depth of the rack. A recommended maximum depth of 525 mm is shown in Figure 3. A power supply unit is not allowed to extend upwards above the maximum height of the Dataway assembly. It should not obstruct the entry or exit of the ventilating air flows in a crate such as that shown in Figure 6. The width of a rear-mounted power supply is limited to 447 mm.

4.2 *Plug-in units*

Basically a plug-in unit consists of a front panel with fixing screw, top and bottom runners that slide in the guides of the crate, and an 86-way Dataway connector plug. The connector plug is typically an integral part of a printed-wiring card, but may be a separate male connector mounted at the rear of the plug-in unit. A plug-in unit may occupy more than one station and, if so, may have more than one set of runners and more than one connector plug.

Unless otherwise indicated, all plug-in units shall conform to Figure 4 and those parts of Figure 5 defining the connector plug.

The following sections are comments on these figures.

4.2.1 *Dimensions*

The horizontal datum of a plug-in unit is the edge of the lower runner. The vertical datum is the rear face of the front panel. The upper and lower parts of the rear face should be in contact with the cross-members of the crate when the plug-in unit is fully inserted. Figure 4 therefore requires that the upper and lower 11 mm of the rear face of the front panel are free from projections, other than the fixing screws.

Figure 4 shows the dimensions of single-width and double-width plug-in units and gives general formulae for the front-panel widths of units.

It is recommended that the fixing screw should also provide a jacking action to assist in overcoming the insertion and withdrawal forces of the connector socket. The fixing screw of a single-width plug-in unit is located on the centre line of the front panel. If a multiple-width unit has only one fixing screw, and this has a jacking action, the screw should be positioned to give the most effective assistance against the insertion and withdrawal forces of the Dataway connector or connectors (hence it should be at the same station as a single connector or approximately symmetrical with respect to two or more connectors).

Above the maximum height of the Dataway assembly, there may be projections at the rear of the plug-in unit, extending more than 290 mm from the vertical datum. Below this height, in order to provide clearance for the connector socket, only the connector plug is allowed to extend beyond 290 mm.

There should be adequate ventilation through the bottom and top of each plug-in unit to remove any heat generated within the unit.

4.2.2 Dataway connector plug

The dimensions of the connector plug are shown in Figures 5.1, 5.2 and 5.3.

The full 86 contacts are always present and extend to the extreme edge of the plug, without a chamfer, in order to avoid the risk of damage to the contact plating of connector sockets by exposed abrasives in the substrate of the connector plug.

Chamfers are provided at the top and bottom of the connector socket and are therefore not needed at the top and bottom corners of the connector plug where the maximum permitted chamfer is 1 mm × 1 mm. For at least 13 mm from the edge of the plug, the contacts are straight and plated.

The dimensions of the contacts of the connector plug are shown in Figure 5.3. The position of each edge is defined by a dimension (h , H) relative to the horizontal datum and is completely independent of the position of all other edges on both sides of the plug. The lowest contact on each side of the plug may be extended to the horizontal datum in order to reduce the impedance of the 0 V line.

4.2.3 Insertion of the plug-in unit into the crate

In the initial stages of insertion, the plug-in unit is supported by the lower guide in the crate. The upper runner, although within the guide, has some vertical clearance. When the plug-in unit is fully inserted, the connector plug is located by the connector socket and the front panel is supported by the fixing screw. The top and bottom runners are then within the guides and approximately parallel to them, but both have some vertical clearance. The transition between these two states is described in detail below.

The dimensions of the guides and runners (see Figures 1 and 4) ensure that the plug-in unit moves freely and is guided so that the leading edge of the connector plug enters the chamfers of the connector socket. The lower corner of the leading edge of the plug comes into contact with the chamfer at the bottom of the connector socket. Further insertion of the plug-in unit lifts the connector plug until its lower edge rests on the horizontal datum face of the connector socket. Even a connector plug with the maximum permitted 1 mm × 1 mm chamfer will have been lifted into correct alignment before any electrical contact occurs between the connector plug and socket. The position of maximum insertion without electrical contact, even with a maximum thickness plug, is defined in Figure 5.5 with respect to the vertical datum of the connector socket.

Before this point has been reached, it will have been possible to engage the fixing screw in the corresponding tapped hole in the lower cross-member of the crate. This can be facilitated by having a tapered end to the screw, so that the front panel is lifted into the correct alignment. The fixing screw has a jacking action which may be used to assist the plug-in unit further into the crate.

Further insertion of the plug-in unit brings the contacts of the plug and socket into engagement, and the insertion force of the connector is encountered. The recommended maximum insertion and withdrawal forces are 80 N for each connector plug. Forces in excess of this may cause difficulty in inserting and withdrawing the plug-in unit and may also result in damage.

Figure 5.5 defines, with respect to the vertical datum of the connector socket, the line beyond which there is reliable contact between corresponding contacts on the plug and socket, even with a plug of minimum thickness.

Finally, when the plug-in unit is fully inserted in the crate, the leading edge of the connector plug is nominally at the vertical datum of the connector socket and the lower datum face of the front panel of the plug-in unit is in contact with the lower cross-member of the crate. However, the forces due to the connector socket and jacking screw are not in line and tend to lift the connector plug off the horizontal datum of the socket, in which case there may be clearance between the upper datum face of the front panel and the upper cross member. Figure 5.5 ensures that there is adequate clearance beyond the extreme position of the connector plug, by defining a minimum distance between the vertical datum of the socket and any internal obstruction.

4.2.4 Printed-wiring card

Figure 8 gives recommended dimensions for a printed-wiring card suitable for use with typical (but not necessarily all) commercially available frameworks for plug-in units conforming to this specification.

4.2.5 Other connectors

Connectors or other components such as switches may be mounted on the front panel, or at the rear of the plug-in unit above the maximum height limit of the Dataway assembly.

Coaxial connectors are to be chosen among connectors of the following characteristics:¹⁾

- miniature,
- impedance 50 Ω ,
- snap on.

4.3 Adaptor for NIM units

Plug-in units conforming to type N module of IEC Publication 482, Dimensions of Electronic Instrument Modules (for Nuclear Electronic Instruments) (NIM unit) can be inserted into the guides of a CAMAC crate. In order to supply power to a NIM unit, which is shorter than a CAMAC plug-in unit, an adaptor is required between the Dataway connector socket and the connector on the NIM unit. The essential dimensions of such an adaptor are given in Figure 7.

4.4 The Dataway

Communication between plug-in units takes place through the Dataway. This passive multi-wire highway is incorporated in the crate and links the Dataway connector sockets at all stations. The Dataway consists of signal lines and power lines, as shown in Table I.

The extreme right-hand station, as viewed from the front of the crate, has the special role of "control station". The data lines in the Dataway are accessible at the remaining "normal stations", but not at the control station.

The assignment of contacts at the Dataway connector and their connections to bus-lines, individual lines and patch contacts shall conform to Table II for normal stations and Table III for the control station. The control station shall be to the right of all normal stations.

The method of construction of the Dataway must be consistent with the signal standards for signal lines (see Clause 7) and with the maximum current loads specified for the power lines (see Clause 8).

Most signal lines are "bus-lines" linking corresponding contacts of the Dataway connector sockets at all normal stations and, in some cases, the control station. There are also "individual lines", each linking one contact at a normal station to one contact at the control station. At each station there are contacts for unspecified uses. Two of these contacts are linked across all normal stations to form "free bus-lines". The remainder are available as "patch contacts", but do not have specified Dataway wiring. The Dataway construction may extend these patch contacts, and others associated with the individual lines and certain bus-lines, to more readily accessible "patch points" to which patch connections can be attached.

The power lines link corresponding contacts of the Dataway connector sockets at all stations. The power return line (0 V) links two contacts in parallel at all stations.

Apart from this, the construction of the Dataway is not specified. Appropriate techniques include printed wiring on flexible or rigid substrates (with and without ground planes), and soldered or wrapped wiring. Particular attention should be given to the cross-coupling between signal lines, and to their capacitance to ground. Relatively high voltages are encountered on three power lines (+200 V d.c., 117 V a.c. live, and 117 V a.c. neutral).

¹⁾ An example of a recommended connector is given in document EUR 4100e (1972). There may, however, be special circumstances requiring the use of other connectors in order to suit a specific external equipment with which the plug-in unit is closely associated.

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TABLE I
Standard Dataway usage

Title	Designation	Number of contacts	Use at a module
Command			
Station number	N	1	Selects the module (individual line from control station).
Sub-address	A1, 2, 4, 8	4	Selects a section of the module.
Function	F1, 2, 4, 8, 16	5	Defines the function to be performed in the module.
Timing			
Strobe 1	S1	1	Controls first phase of operation (Dataway signals must not change).
Strobe 2	S2	1	Controls second phase (Dataway signals may change).
Data			
Write	W1 to W24	24	Bring information to the module.
Read	R1 to R24	24	Take information from the module.
Status			
Look-at-me	L	1	Indicates request for service (individual line to control station).
Busy	B	1	Indicates that a Dataway operation is in progress.
Response	Q	1	Indicates status of feature selected by command.
Command accepted	X	1	Indicates that module is able to perform action required by the command.
Common controls			<i>Operate on all features connected to them, no command required.</i>
Initialize	Z	1	Sets module to a defined state (accompanied by S2 and B).
Inhibit	I	1	Disables features for duration of signal.
Clear	C	1	Clears registers (accompanied by S2 and B).
Non-standard connection			
Free bus-lines	P1, P2	2	For unspecified uses.
Patch contacts	P3 to P5	3	For unspecified interconnections. No Dataway lines.
Mandatory power lines			<i>The crate is wired for mandatory and additional lines.</i>
+24 V d.c.	+24	1	
+6 V d.c.	+6	1	
-6 V d.c.	-6	1	
-24 V d.c.	-24	1	
0 V	0	2	Power return.
Additional power lines			<i>These lines are reserved for the following power supplies.</i>
+200 V d.c.	+200	1	
+12 V d.c.	+12	1	Low current for indicators, etc.
-12 V d.c.	-12	1	
117 V a.c. (Live)	ACL	1	
117 V a.c. (Neutral)	ACN	1	
Clean earth	E	1	Reference for circuits requiring clean earth.
Reserved	Y1, Y2	2	Reserved for future allocation.
Total		86	

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TABLE II
Contact allocation at a normal station
(Viewed from front of crate)

Bus-line	Free bus-line	P1	B	Busy	Bus-line
Bus-line	Free bus-line	P2	F16	Function	Bus-line
Individual patch contact		P3	F8	Function	Bus-line
Individual patch contact		P4	F4	Function	Bus-line
Individual patch contact		P5	F2	Function	Bus-line
Bus-line	Command accepted	X	F1	Function	Bus-line
Bus-line	Inhibit	I	A8	Sub-address	Bus-line
Bus-line	Clear	C	A4	Sub-address	Bus-line
Individual line	Station number	N	A2	Sub-address	Bus-line
Individual line	Look-at-me	L	A1	Sub-address	Bus-line
Bus-line	Strobe 1	S1	Z	Initialize	Bus-line
Bus-line	Strobe 2	S2	Q	Response	Bus-line
24 write bus-lines		W24	W23		
		W22	W21		
		W20	W19		
		W18	W17		
		W16	W15		
		W14	W13		
		W12	W11		
		W10	W9		
		W8	W7		
		W6	W5		
		W4	W3		
		W2	W1		
24 read bus-lines		R24	R23		
		R22	R21		
		R20	R19		
		R18	R17		
		R16	R15		
		R14	R13		
		R12	R11		
		R10	R9		
		R8	R7		
		R6	R5		
		R4	R3		
		R2	R1		
Power bus-lines		-12	-24	-24 V d.c.	Power bus-lines
		+200	-6	-6 V d.c.	
		ACL	ACN	117 V a.c. neutral	
		Y1	E	Clean earth	
		+12	+24	+24 V d.c.	
		Y2	+6	+6 V d.c.	
		0	0	0 V (power return)	

5. Use of the Dataway lines

Each line of the Dataway shall be used in accordance with the mandatory requirements detailed in the following sub-clauses and summarized in Table I.

A typical Dataway operation involves at least two plug-in units, one of which acts as a "controller" and the other as a controlled "module" (see Sub-clause 3.2).

There are two types of Dataway operations: command operations and unaddressed operations. During command operation, the controller generates a "command" consisting of signals on individual "station number" lines to specify one or more modules, on the "sub-address" bus-lines to specify a sub-section of the module, and on the "function" bus-lines to specify the operation to be performed.

During unaddressed operations, there is no command, but the controller generates one of the "common control" signals on the "initialize" or "clear" bus-lines, and this operates on all modules connected to the bus-line.

During command operations and unaddressed operations, the controller generates a signal on the "busy" bus-line. The busy signal is available at all stations to indicate that a Dataway operation is in progress.

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TABLE III
Contact allocation at the control station
(Viewed from front of crate)

Individual patch contact		P1	B	Busy	Bus-line
Individual patch contact		P2	F16	Function	Bus-line
Individual patch contact		P3	F8	Function	Bus-line
Individual patch contact		P4	F4	Function	Bus-line
Individual patch contact		P5	F2	Function	Bus-line
Bus-line	Command accepted	X	F1	Function	Bus-line
Bus-line	Inhibit	I	A8	Sub-address	Bus-line
Bus-line	Clear	C	A4	Sub-address	Bus-line
Individual patch contact		P6	A2	Sub-address	Bus-line
Individual patch contact		P7	A1	Sub-address	Bus-line
Bus-line	Strobe 1	S1	Z	Initialize	Bus-line
Bus-line	Strobe 2	S2	Q	Response	Bus-line
24 individual look-at-me lines		L24	N24	24 individual station number lines	
		L23	N23		
		L22	N22		
		L21	N21		
		L20	N20		
		L19	N19		
		L18	N18		
		L17	N17		
		L16	N16		
		L15	N15		
		L14	N14		
		L13	N13		
		L12	N12		
		L11	N11		
		L10	N10		
		L9	N9		
		L8	N8		
		L7	N7		
		L6	N6		
		L5	N5		
		L4	N4		
		L3	N3		
		L2	N2		
		L1	N1		
Power bus-lines	-12 V d.c.	-12	-24	-24 V d.c.	Power bus-lines
	+200 V d.c.	+200	-6	-6 V d.c.	
	117 V a.c. live	ACL	ACN	117 V a.c. neutral	
	Reserved	Y1	E	Clean earth	
	+12 V d.c.	+12	+24	+24 V d.c.	
	Reserved	Y2	+6	+6 V d.c.	
	0 V (power return)	0	0	0 V (power return)	

Two timing signals, strobes S1 and S2, are generated in sequence on separate bus-lines during command operations. Only strobe S2 is mandatory during unaddressed operations, but S1 may also be generated.

During a Dataway command, operation there may be a read data transfer from a module to the controller, or a write data transfer from the controller to a module, or neither.

In response to a read command, the addressed module establishes read data signals which are available to the controller from the time of strobe S1 onwards. In response to a write command, the addressed module accepts write data signals from the controller at the time of strobe S1.

The addressed module indicates by a signal on the "command accepted" bus-line whether it is able to perform the action required by the command. It may also transmit one bit of status information on the "response" bus-line. The controller accepts the command accepted and response signals at the time of strobe S1.

Any module may generate a signal on its individual "look-at-me" line to indicate that it requires attention.

The use of each Dataway line is defined in the following sub-clause. The relationship between signals in order to generate specific commands is defined in Clause 6 and the electrical signal standards, including timing, are defined in Clause 7.

The sequence of events during command operations is described in Sub-clause 7.1.3.1 and shown in Figure 9. The sequence during unaddressed operations is described in Sub-clause 7.1.3.2 and shown in Figure 10.

5.1 Commands

A command consists of signals on:

- the individual station number lines (specifying a module or modules);
- the four sub-address lines (specifying a sub-section of the module);
- the five function lines (specifying the type of operation).

The command signals are maintained for the full duration of the Dataway operation. They are accompanied by a signal on the busy bus-line, which indicates to all units that a Dataway operation is in progress.

Plug-in units shall not rely on the state of signals on the sub-address and function lines when no command operation is in progress.

5.1.1 Station number (*N*)

Each normal station is addressed by a signal on an individual station number line (N_i) which comes from a separate contact at the control station (see Tables II and III). The stations are numbered in decimal from the left-hand end as viewed from the front, beginning with station 1 (addressed by N_1).

There is no restriction on the number of stations that can be addressed simultaneously.

5.1.2 Sub-address (*A8, A4, A2, A1*)

Different sub-sections of a module are addressed by signals on the 4 A bus-lines. These signals are decoded in the module to select one of up to 16 sub-addresses, numbered in decimal from A(0) to A(15).

The sub-address may be used to select, for example, a register within the module, or a feature that is to control the Response signal (Q), or a section of the module that is to be operated on by functions such as Enable, Disable, and Execute. The use made of the sub-address within a module is discussed in relation to the function codes in Clause 6.

Each sub-address code used in a module shall be fully decoded in the module. Full decoding means that all 4 Dataway sub-address signals are used in the decoding process.

The sub-address codes are designated A(0), A(1), A(2), A(3), etc., to distinguish them from the individual sub-address lines A1, A2, A4 and A8. For example, the sub-address signals $A_1 = 1$, $A_2 = 1$, $A_4 = 0$ and $A_8 = 0$ represent the code A(3).

5.1.3 Function (*F16, F8, F4, F2, F1*)

The function to be performed at the specified sub-address in the selected module or modules is defined by the signals on the 5 F bus-lines. These signals are decoded in the module to select one of up to 32 functions, numbered in decimal from F(0) to F(31). The definitions of the 32 function codes are summarized in Table IV and are detailed in Sub-clause 6 in relation to the command structure.

The function codes are subdivided into three groups, involving read operations, write operations and operations with no transfer of data: the "standard" function codes having defined actions in modules and controllers, the "reserved" function codes for any future additions to the standard codes and the "non-standard" function codes whose use is neither defined in detail nor co-ordinated by standards.

Each function code used in a module shall be fully decoded in the module. Full decoding means that all 5 Dataway function signals are used in the decoding process.

The function codes are designated F(0), F(1), F(2), F(3), etc., to distinguish them from the individual function lines F1, F2, etc. For example, the function signals $F_1 = 1$, $F_2 = 0$, $F_4 = 0$, $F_8 = 1$ and $F_{16} = 1$ represent the code F(25).

TABLE IV
The function codes

Code F()	Function	Use of R and W lines	Function signals					Code F()
			F16	F8	F4	F2	F1	
0	Read group 1 register	Functions using the R lines	0	0	0	0	0	0
1	Read group 2 register		0	0	0	0	1	1
2	Read and clear group 1 register		0	0	0	1	0	2
3	Read complement of group 1 register		0	0	0	1	1	3
4	Non-standard	Functions using the R lines	0	0	1	0	0	4
5	Reserved		0	0	1	0	1	5
6	Non-standard		0	0	1	1	0	6
7	Reserved		0	0	1	1	1	7
8	Test look-at-me	Functions not using the R or W lines	0	1	0	0	0	8
9	Clear group 1 register		0	1	0	0	1	9
10	Clear look-at-me		0	1	0	1	0	10
11	Clear group 2 register		0	1	0	1	1	11
12	Non-standard	Functions not using the R or W lines	0	1	1	0	0	12
13	Reserved		0	1	1	0	1	13
14	Non-standard		0	1	1	1	0	14
15	Reserved		0	1	1	1	1	15
16	Overwrite group 1 register	Functions using the W lines	1	0	0	0	0	16
17	Overwrite group 2 register		1	0	0	0	1	17
18	Selective set group 1 register		1	0	0	1	0	18
19	Selective set group 2 register		1	0	0	1	1	19
20	Non-standard	Functions using the W lines	1	0	1	0	0	20
21	Selective clear group 1 register		1	0	1	0	1	21
22	Non-standard		1	0	1	1	0	22
23	Selective clear group 2 register		1	0	1	1	1	23
24	Disable	Functions not using the R or W lines	1	1	0	0	0	24
25	Execute		1	1	0	0	1	25
26	Enable		1	1	0	1	0	26
27	Test status		1	1	0	1	1	27
28	Non-standard	Functions not using the R or W lines	1	1	1	0	0	28
29	Reserved		1	1	1	0	1	29
30	Non-standard		1	1	1	1	0	30
31	Reserved		1	1	1	1	1	31

5.2 Strobe signals (*S1* and *S2*)

Both strobe signals shall be generated by the controller during each command operation.

Plug-in units shall not take irreversible action based on the command or data signals until the time of *S1*. Actions concerned with the acceptance of data and status information from the *R*, *W*, *Q* and *X* lines shall be initiated at the time of *S1*. Other actions may also be timed by *S1*, but shall not change the state of signals on the *R* and *W* lines.

Any actions that can change the state of Dataway read or write signals shall be initiated by the second strobe *S2*. For example, *S2* shall be used if it is required to clear a register whose output is connected to the Dataway.

Strobe *S2* shall be generated during each unaddressed operation to indicate when modules have to accept the common control signal.

Strobe *S1* may also be generated but this is not mandatory and modules cannot rely on it.

5.3 Data

All information carried by the read and write lines is conveniently described as data, although it may be information concerned with status or control features in modules. Information that is transferred to or from a control register in a module is thus regarded as data.

Up to 24 bits may be transferred in parallel between the controller and the selected module. Independent lines are provided for the read and write directions of transfer.

If the bits of a data word have different numerical significance, line R_n should be used for a higher-order bit than R_{n-1} and W_n for a higher-order bit than W_{n-1} .

It is recommended that controllers have 24-bit capability. For particular applications, assemblies are permitted in which the controller has a word length less than 24 bits and the modules have an equal or smaller word length.

Plug-in units shall not rely on the state of signals on the read and write bus-lines when no command operation is in progress.

5.3.1 Write lines (*W1* to *W24*)

The controller generates data signals on the *W* bus-lines during each write operation. The *W* signals shall reach a steady state before *S1* and shall be maintained until the end of the operation, unless modified by *S2*. Strobe *S1* shall be used by the modules to strobe the data unless there are very strong technical reasons for choosing *S2*.

5.3.2 Read lines (*R1* to *R24*)

Data signals are set up on the *R* bus-lines by the module during a read operation. The *R* signals shall reach a steady state before *S1* and shall be maintained for the full duration of the Dataway operation, unless the state of the data source is changed by *S2*. The controller shall initiate action concerned with the acceptance of data from the *R* lines at the time of the Strobe *S1* and shall not take irreversible action before this.

5.4 Status information

Status information is conveyed by signals on the look-at-me (*L*), busy (*B*), response (*Q*) and command accepted (*X*) lines.

5.4.1 Look-at-me (*LAM* or *L*)

The look-at-me lines, like the *N* lines, are individual connections from each normal station to separate contacts at the control station (*L1* from station 1, etc.).

Any module may generate a signal on its individual line (L_i) to indicate that it requires attention. Modules that occupy more than one station may indicate different demands by signals on the appropriate *L* lines.

The L signal generated by a module may represent demands for attention originating from more than one look-at-me source (LAM source) in the module.

Note. — Some LAM structure options are given in document EUR 4100c (1972).

5.4.1.1 *Look-at-me: Clear, disable and test*

Provision shall be made for resetting each bit of the LAM status register individually either by a clear look-at-me operation F(10) (see Sub-clause 6.2.3), or by a selective clear group 2 operation F(23) (see Sub-clauses 5.4.1.2 and 6.3.6). All LAM status bits shall be reset collectively by the initialize signal (see Sub-clause 5.5.1).

If the LAM request calls for some specific action (for example, reading the contents of a data register) then the corresponding bit of the LAM status register should also be cleared when the appropriate Dataway operation is performed.

A module that has generated $L = 1$ shall not clear the LAM status register until it receives an appropriate command or the initialize signal.

Each module that generates L should have a means of enabling and disabling the LAM requests. This may be done by loading and clearing a mask register, or by enable and disable commands.

All LAM requests that can be disabled by commands shall also be disabled by the initialize signal (Z).

A module that generates L shall have a means of testing the L signal by a test look-at-me operation (Function code F(8), with a sub-address distinguishing this from tests of individual LAM requests). If there are several LAM sources the corresponding LAM requests shall also be capable of being examined either by test look-at-me operations associated with appropriate sub-addresses or by reading a LAM request pattern in a read operation.

5.4.1.2 *Look-at-me: Commands for access*

Modules may contain registers for LAM information. These registers are not mandatory, but if included they should be accessed as group 2 registers at the following sub-addresses:

- LAM status register A(12);
- LAM mask register A(13);
- LAM request register A(14).

Corresponding bit-positions should be associated with the same LAM source.

The state of each data-bit read from the LAM status or LAM request register is the same as the state of the Q response that would be obtained by a test status or test look-at-me operation.

The data word read from A(12) should have LAM status information in the low-order bits and may also contain other status information. Each bit of the data word loaded into A(13) should be in the "1" state to enable the corresponding LAM request, and in the "0" state to disable it.

The operations used to access LAM information may be divided into two classes. One class consists of read F(1), write F(17), clear F(11), selective set F(19) and selective clear F(23) addressed to the group 2 LAM registers described above. This class is preferred for operations on modules with many LAM sources. The other class consists of clear LAM F(10), enable F(26), disable F(24), test status F(27) and test LAM F(8) addressed to specific demands. This class is preferred for modules with few LAM sources. In the one class a LAM source, LAM(i), is associated with bit position (i) in data words, and in the other class with sub-address A(i). For ease of programming, it is recommended that all the operations related to a particular LAM source should belong to only one class.

5.4.1.3 Look-at-me: Gating

When a module that is generating $L_i = 1$ receives a command that will cause it to cease doing so, it shall inhibit the $L = 1$ signal or the appropriate LAM request. The inhibit condition shall be effective before strobe S1 and shall be maintained until the end of the Dataway operation.

This requirement may be met very simply by inhibiting the L signal output when a module is addressed by any command ($L_i = 0$ when $N_i = 1$). Unaddressed modules can thus initiate $L = 1$ at any time, but addressed modules cannot do so until the end of the current Dataway operation.

The requirements may be met more precisely by inhibiting only those LAM status signals that are to be cancelled by the current command. The ability to initiate $L_i = 1$ during a Dataway operation is thus extended to all LAM requests that are not being cancelled. This requires the recognition of $N_i = 1$ with the specific functions and sub-addresses, and the generation of appropriate inhibit conditions.

The requirement may be interpreted in ways intermediate between these extremes. For example, the ability to initiate $L_i = 1$ can be extended to all LAM requests during most Dataway operations if the L signal output is inhibited when the module is addressed by $N_i = 1$ together with appropriate, easily identifiable groups of functions and sub-addresses.

5.4.2 Busy (B)

The busy signal is used to interlock various aspects of a system that can compete for the use of the Dataway. The signal $B = 1$ indicates to all units that a Dataway operation is in progress.

The busy signal $B = 1$ shall be generated during each Dataway command operation (when N signals are also generated) and during unaddressed operations (when Z or C are generated).

5.4.3 Response (Q)

During every command operation, the addressed module may generate a signal on the Q bus-line to indicate the status of any selected feature of the module.

The controller shall initiate action concerned with the acceptance of the status information from the Q line at the time of strobe S1 and shall not take irreversible action before this.

In read and write operations (see Sub-clauses 6.1 and 6.3), addressed modules shall establish the signal $Q = 0$ or $Q = 1$ before strobe S1 and shall maintain it until at least strobe S2.

In test look-at-me operations (see Sub-clause 6.2.1), addressed modules may initiate the Q signal at any time during a Dataway operation if the status of the appropriate LAM request changes. If $Q = 1$ has been initiated, it will remain static until the end of the operation since this command is not allowed to reset LAM status.

In all operations other than read, write and test look-at-me, the Q signal is permitted to change at any time. There is a risk that status information will be missed if $Q = 1$ is initiated between Strobes S1 and S2 during an operation in which the module resets the status condition at the time of Strobe S2.

In any operation, the Q signal conveys only one bit of information, which should be clearly defined for each sub-address and function code used by the module.

Examples of the use of the Q signal during read and write operations are given in Sub-clauses 5.4.3.1, 5.4.3.2 and 5.4.3.3, which define three methods of transferring blocks of data. These three methods are summarized in the following table. However, the status information transmitted by the addressed module during read and write operations is not restricted to these examples.

Response	Q mode		
	Address scan	Repeat	Stop
$Q = 1$	Register present	Register ready	Within block
$Q = 0$	Register absent	Register not ready	End of block

5.4.3.1 Address scan mode

If a module contains registers that are intended to be accessed sequentially in address scan mode, they shall be located at consecutive sub-addresses starting at $A(0)$. During read and write operations, the module shall generate $Q = 1$ at all sub-addresses at which these registers are present and $Q = 0$ at the first unoccupied sub-address, if any. A module with n such registers shall therefore generate $Q = 1$ at $A(0)$ to $A(n - 1)$. If $n < 16$, the module shall generate $Q = 0$ at $A(n)$.

5.4.3.2 Repeat mode

If a module contains a register that is intended to be accessed in repeat mode, it shall generate $Q = 1$ during a read or write operation if the register is ready to participate in a data transfer. It shall generate $Q = 0$ during a read or write operation if the register is not ready to participate in a data transfer.

5.4.3.3 Stop mode

If a module contains a register that is intended to be accessed in stop mode, it shall generate $Q = 1$ during each read or write operation while the block of data is being transferred and shall generate $Q = 0$ during any further operations after the end-of-block condition has been encountered.

5.4.4 Command accepted (X)

Whenever a module is addressed during a command operation, it shall generate $X = 1$ on the command accepted bus-line if it recognizes the command as one that it is equipped to perform, either within the module or in association with external equipment. The signal on the X line shall reach a steady state before $S1$ and shall be maintained until $S2$. Controllers that accept the X signal shall do so at the time of $S1$.

The signal $X = 0$ should indicate a serious malfunction, for example a module that is not present, not powered, lacks external connections or is not equipped to perform the required action. The action taken by the controller in response to $X = 0$ may, for example, be to call for intervention by the operator or operating system.

5.5 Common controls (Z , C , I)

Three common control signals are available at all stations, without requiring addressing by a command, in order to initialize all units (typically after switch-on), to clear data registers, or to inhibit features such as data taking.

During unaddressed Dataway operations, either initialize (Z) or clear (C) is generated by the controller and received by each unit connected to the appropriate bus-line.

The common control signal inhibit (I) is not associated with Dataway operations. It may be generated at any time and is received by each unit connected to the I bus-line.

The initialize (Z) and clear (C) signals shall be accompanied by busy (B) and strobe $S2$ signals, in a timing sequence as described in Sub-clause 7.1.3.2 and shown in Figure 10. The sequence is permitted to include strobe $S1$, but units shall not rely on the generation of $S1$ with Z or C .

5.5.1 Initialize (Z)

The initialize signal is intended to be used during system start-up.

Initialize shall have absolute priority over other signals. In response to $Z = 1$, all data and control registers shall be set to a defined initial state, all LAM status registers shall be reset and, if possible, all LAM requests shall be disabled.

Units that generate Z shall also initiate a sequence including busy, strobe $S2$ and inhibit (see Sub-clause 5.5.2).

Units that accept Z shall gate it with strobe $S2$ as a protection against spurious signals on the Z line.

5.5.2 Inhibit (I)

The signal $I = 1$ shall inhibit any feature to which it is connected in a module.

When any unit generates the initialize signal $Z = 1$, it shall also generate $I = 1$. The inhibit signal accompanying Z shall be established by time t_1 (see Figure 10) and shall be maintained for at least the duration of the Z signal. All units that generate I and can maintain $I = 1$ shall respond to Z - $S2$ by generating and maintaining $I = 1$ until specifically reset.

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5.5.3 Clear (C)

The signal C = 1 shall clear all registers and bistables to which it is connected.

Units that generate C shall also initiate a sequence including busy and strobe S2.

Units that accept C shall gate it with strobe S2 as a protection against spurious signals on the C line.

5.6 Non-standard connections (P1 to P7)

Five contacts (P1 to P5) on the Dataway connector at each normal station, and seven contacts (P1 to P7) at the control station, are available for unspecified uses.

5.6.1 Free bus-lines (P1, P2)

The contacts P1 and P2 at all normal stations shall be linked by two free bus-lines.

Each plug-in unit is permitted to generate signals on either or both of these lines, or to accept signals from them. Within the plug-in unit there shall be means by which any access to these lines can be disconnected or disabled.

Signals on the free bus-lines shall either conform to Sub-clause 7.1.4 and Table VII (with distributed pull-ups and freedom to vary the number of inputs and outputs) or to Sub-clause 7.1.2 and Table VI (with one pull-up on each line, not necessarily in the controller, and with current standards as for read or write lines).

5.6.2 Individual patch contacts

Contacts P3 to P5 at each normal station and P1 to P7 at the control station are not wired to Dataway lines. They are available for patch connections to other of these contacts, to optional patch points on certain Dataway lines, to the 0 V line, or to external equipment.

Patch connections shall not be essential for the operation of the main features of general purpose units.

The signals on patch connections shall conform with Sub-clause 7.1.4 and Tables V and VII.

5.7 Power lines

The Dataway shall include lines for all the mandatory, additional and reserved power supplies shown in Table I.

Two reserved power lines (Y1, Y2) shall not be used until they have been fully defined by the ESONE Committee, for example, to meet the needs of any new circuit technology for which the existing supply voltages are unsuitable.

Details of the voltage tolerances and permitted loadings of the power lines are given in Clause 8.

6. Dataway commands

During a Dataway command operation, modules and controllers shall perform the actions specified for the particular command. They may also perform additional internal actions, but these shall not involve transferring to or from the Dataway any data or status information other than that specified for the command. Additional internal actions shall not convert one standard command into another standard command.

The following sub-clauses define the mandatory actions by modules and controllers in response to each command, including mandatory transfers of data and status information via the Dataway. The term "register" is used here and in the summary of the function codes in Table IV to indicate an addressable data source or receiver, without necessarily implying that it has a data storage property.

The function codes F(0) to F(3), F(9), F(11), F(16) to F(19), F(21) and F(23) allow the registers in a module to be divided into two distinct sets, known as group 1 and group 2, so that it is possible to operate on two sets of 16 registers. Within each group the appropriate register is selected by the sub-address. Information concerning status or system organization, or requiring restricted access, should be held in group 2 registers (for example, see Sub-clause 5.4.1.2).

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If a module allows a descriptor (module characteristic) to be read, the command used should be: read group 2 register, F(1). A(15).

6.1 *Read commands: Function codes F(0) to F(7)*

Read commands are identified by the combination $F16 = 0$ and $F8 = 0$ in the function code. All read commands involve the transfer of data and status information from a module to the controller via the R, Q and X lines (see mandatory statements in Sub-clauses 5.3.2, 5.4.3 and 5.4.4).

Standards for the use of the Q signal in read operations are given in Sub-clause 5.4.3.

6.1.1 *Read group 1 register: Code F(0)*

This command transfers to the controller the contents of a register in the first group in the module. The contents of the register are not changed.

6.1.2 *Read group 2 register: Code F(1)*

This command transfers to the controller the contents of a register in the second group in the module. The contents of the register are not changed.

6.1.3 *Read and clear group 1 register: Code F(2)*

This command transfers to the controller the contents of a register in the first group in the module. The contents of the register are cleared at time S2.

6.1.4 *Read complement of group 1 register: Code F(3)*

This command transfers to the controller the “ones” complement of the contents of a register in the first group in the module. The contents of the register are not changed.

The command is provided mainly as a means of error detection. A read transfer with F(0) or F(2) can be checked by preceding it with a transfer from the same register with F(3). The two data words received by the controller should be complementary. A write transfer with F(16) can be checked by following it with a read transfer from the module with F(3). The data words sent and received by the controller should be complementary.

6.1.5 *Other read commands: Codes F(4) to F(7)*

These commands transfer the contents of a register in the module to the controller. Codes F(4) and F(6) are available as non-standard functions. Codes F(5) and F(7) are reserved for extensions of the standard functions.

6.2 *Control commands: Function codes F(8) to F(15)*

This first group of control commands is identified by $F8 = 1$ and $F16 = 0$ in the function code. Information is not transferred on either the R or W lines. However, status information may be conveyed on the Q line in response to any of these commands. The signal on the Q line is permitted to change at any time. It is strobed into the controller at time S1 and may, except in operations with code F(8), be reset by strobe S2. There is a risk that information can be lost due to Q signals appearing between S1 and S2.

6.2.1 *Test look-at-me: Code F(8)*

This command transfers to the controller a signal on the response line Q, representing the state of the L signal or a LAM request in the module (see Sub-clauses 5.4.1 and 5.4.1.1). The response shall be $Q = 0$ if the feature is in the “0” state or is prevented, by masking or gating, from contributing to a “1” state L signal. The LAM status shall not be reset by this command.

6.2.2 *Clear group 1 register: Code F(9)*

This command clears the contents of a register in the first group in the module.

6.2.3 *Clear look-at-me: Code F(10)*

This command resets a LAM status in the module (see Sub-clause 5.4.1).

6.2.4 Clear group 2 register: Code F(11)

This command clears the contents of a register in the second group in the module.

6.2.5 Other control commands: Codes F(12) to F(15)

These commands do not transfer data on the R or W bus-lines. Codes F(12) and F(14) are available for use as non-standard functions. Codes F(13) and F(15) are reserved for extensions to the standard functions.

6.3 Write commands: Function codes F(16) to F(23)

Write commands are identified by the combination $F_{16} = 1$ and $F_8 = 0$ in the function code. All write commands involve the transfer of data from the controller to a module via the write bus-lines, and status information from a module to the controller via the Q and X lines (see mandatory statements in Sub-clauses 5.3.1, 5.4.3 and 5.4.4).

Standards for the use of the Q signal in write operations are given in Sub-clause 5.4.3.

6.3.1 Overwrite group 1 register: Code F(16)

This command forces each bit of a register in the first group in the module to the same state as the corresponding data bit transmitted by the controller.

The effect of this command is to write data bit W_i into bit M_i of the group 1 register.

Thus: $M_i = W_i$

6.3.2 Overwrite group 2 register: Code F(17)

This command forces each bit of a register in the second group in the module to the same state as the corresponding data bit transmitted by the controller.

The effect of this command is to write data bit W_i into bit M_i of the group 2 register.

Thus: $M_i = W_i$

6.3.3 Selective set group 1 register: Code F(18)

This command operates on selected bit positions of a register in the first group in the module. The bit positions are selected by "1" bits in a data word transmitted by the controller, and their contents are set to the "1" state. The contents of unselected bit positions are unchanged.

The effect of this command is to form the inclusive OR function of the data bit W_i and the bit M_i in the group 1 register.

Thus: $M_i = W_i + M_i$

6.3.4 Selective set group 2 register: Code F(19)

This command operates on selected bit positions of a register in the second group in the module. The bit positions are selected by "1" bits in a data word transmitted by the controller, and their contents are set to the "1" state. The contents of unselected bit positions are unchanged.

The effect of this command is to form the inclusive OR function of the data bit W_i and the bit M_i in the group 2 register.

Thus: $M_i = W_i + M_i$

6.3.5 Selective clear group 1 register: Code F(21)

This command operates on selected bit positions of a register in the first group in the module. The bit positions are selected by "1" bits in a data word transmitted by the controller, and their contents are cleared to the "0" state. The contents of unselected bit positions are unchanged.

The effect of this command is to form the function of the data bit W_i and the bit M_i in the group 1 register:

$$M_i = \overline{W_i} \cdot M_i$$

6.3.6 *Selective clear group 2 register: Code F(23)*

This command operates on selected bit positions of a register in the second group in the module. The bit positions are selected by "1" bits in a data word transmitted by the controller, and their contents are cleared to the "0" state. The contents of unselected bit positions are unchanged.

The effect of this command is to form the function of the data bit W_i and the bit M_i in the group 2 register:

$$M_i = \overline{W_i} \cdot M_i$$

6.3.7 *Other write commands: Codes F(20) and F(22)*

These codes are available for use as non-standard functions that operate on some or all bits of a register in the module in accordance with the data transmitted by the controller.

6.4 *Control commands: Function codes F(24) to F(31)*

This second group of control commands is identified by the combination $F8 = 1$ and $F16 = 1$ in the function code. Information is not transferred on either the R or W bus-lines. However, status information may be conveyed on the Q line in response to any of these commands. The signal on the Q line is permitted to change at any time. It is strobed into the controller at time S1 and may, except in operations with code F(27), be reset by strobe S2. There is a risk that information can be lost due to Q signals appearing between S1 and S2.

6.4.1 *Disable: Code F(24)*

This command disables a feature of the module or masks off a signal. The action is initiated by strobe S1 or S2.

6.4.2 *Execute: Code F(25)*

This command initiates or terminates an action when enable or disable is not appropriate. The initiation or termination occurs at the time of strobe S1 or S2. The execute command shall not be used to set a feature of the module that requires a disable command F(24) to reset it, nor to reset a feature that requires an enable command F(26) to set it.

6.4.3 *Enable: Code F(26)*

This command activates or enables a feature of the module or unmask a signal. The action is initiated by strobe S1 or S2.

6.4.4 *Test status: Code F(27)*

This command produces on the Q line a response corresponding to the status of a feature of the module. The feature, which is selected by the sub-address, may be a LAM status but shall not be a LAM request or L signal (use test look-at-me, Sub-clause 6.2.1). The feature shall not be reset by the test status command.

6.4.5 *Other control commands: Codes F(28) to F(31)*

These commands do not transfer information on the R or W bus-lines. Codes F(28) and F(30) are available for use as non-standard functions. Codes F(29) and F(31) are reserved for extensions to the standard functions.

6.5 *External representation of the command*

A command is represented on the Dataway by the 5-bit function code, the 4-bit sub-address code, and signals on the appropriate N lines. This publication does not define the form in which the command should be transmitted externally (e.g. between a computer and a crate). It will generally be convenient to use the same function and sub-address codes.

7. *Signal standards*

The standards specified in this clause apply to signals into and out of plug-in units through:

- a) the Dataway (including timing standards for the main signals associated with Dataway operations);

- b) non-standard connections (P1 to P7) via the Dataway connector.
- c) other connectors on the front panel or at the rear of the unit above the Dataway (with separate standards for terminated and unterminated digital signals and for analogue signals).

The signal standards do not restrict the freedom of designers to use other signals or conventions within units.

7.1 Digital signals on the Dataway

The potentials for the binary digital signals on the Dataway lines have been defined to correspond with those for compatible current sinking logic devices (e.g. the DTL and TTL series). The signal convention has, however, been chosen to be negative logic. The high state (more positive potential) corresponds to logic "0" and the low state (near ground potential) corresponds to logic "1". Intrinsic OR outputs are thus available from standard product ranges.

It is an essential feature of the Dataway that many units may have signal outputs connected to the read, command accepted and response lines. Outputs onto these lines therefore require intrinsic OR gates. The same principle is extended to other lines (command, write, etc.) in order to allow more than one controller-like unit in a crate.

Signal outputs from all plug-in units onto all Dataway lines shall be delivered through intrinsic OR gates. Each line shall be provided with an individual pull-up current source to restore the line to the "0" state in the absence of an applied "1" signal.

The rise and fall times at signal outputs to Dataway lines shall be not less than 10 ns, in order that cross-coupling of signals on the Dataway is not excessive.

7.1.1 Voltage levels for Dataway signals

All Dataway signals shall conform to the voltage levels shown in Table V.

TABLE V
Voltage levels of Dataway signals

	"0" state	"1" state
Accepted at input	+2.0 V to +5.5 V	0 to +0.8 V
Generated at output	+3.5 V to +5.5 V	0 to +0.5 V

7.1.2 Current standards for Dataway signals

All Dataway signals shall conform to the standards for input and output currents shown in Table VI.

As shown also in this table, pull-up current sources for all standard Dataway bus-lines are located in the controller so as to ensure that there is one and only one source per line. The pull-up current sources for the N lines are located in the unit generating the signals and for the L lines in a unit receiving the signals so that the individual lines may be joined or grouped within these units if desired.

The strobe signals S1 and S2, which time all actions in modules, have larger pull-up currents than other signals in order to give improved transition times and immunity against cross-coupling from other Dataway lines.

TABLE VI

Standards for signal currents through Dataway connectors and for pull-up current sources

Where appropriate, the current passing through the Dataway connector of a plug-in unit is defined as a function of the width of the unit ("s" stations). Values are given, as examples, for typical controllers ($s = 2$, control station and one normal station) and other units ($s = 1$).

Note 1. — Although only the controller and one module are connected directly to each N and L line, additional units may be connected via Dataway patch points or auxiliary connectors.

Designation of Dataway signal line	N	L	Q, R, X	W, A, F, B, Z, C, I	S1, S2
Line in "1" state at +0.5 V Minimum current sinking capability (current drawn from line) of each unit generating the signal	6.4 mA	16 mA	Controllers 1.6 (25-s) mA 36.8 mA typical Other units 9.6 + 1.6(25-s) mA 48.0 mA typical 58 + 1.6 (25-s) mA 96.4 mA typical		
Line in "1" state at +0.5 V Maximum current fed into line by each unit receiving the signal	3.2 mA each unit 6.4 mA total (Note 1)	Unit with pull-up current source: 11.2 mA Units without pull-up current source 1.6 mA each: 4.8 mA total (Note 1)	1.6 s mA		
Line in "0" state at +3.5 V Minimum pull-up capability (current fed into line) of the unit with pull-up current source	100 (25-s) μ A 2.3 mA typical for controllers 2.4 mA typical for other units				9.9 mA
Line in "0" state at +3.5 V Maximum current drawn from line by each unit without pull-up current source	200 μ A	100 s μ A			
Location of pull-up current source	Unit generating the signal	One unit receiving the signal	Controller		
Pull-up current I_p from positive potential Line in "1" state at +0.5 V	$6 \text{ mA} \leq I_p \leq 9.6 \text{ mA}$				$38 \text{ mA} \leq I_p \leq 58 \text{ mA}$
Pull-up current I_p from positive potential Line in "0" state at +3.5 V	$2.5 \text{ mA} \leq I_p$				$10 \text{ mA} \leq I_p$

7.1.3 Timing of Dataway signals

The sequence of events during a Dataway command operation is shown in Figure 9 by means of simplified signal waveforms. Sub-clause 7.1.3.1 is an explanation of Figure 9.

The sequence of events during a Dataway unaddressed operation is shown in Figure 10, and an explanation is given in Sub-clause 7.1.3.2.

In both figures the shaded areas indicate the permitted variation in the timing of each signal. The vertical edge of each shaded area corresponds to an ideal signal without delay. The sloping edge corresponds to a signal that reaches the appropriate threshold (0.8 V or 2.0 V) after the maximum permitted delay.

The performance of all plug-in units and Dataway assemblies shall be consistent with the timing requirements shown in Figures 9 and 10.

7.1.3.1 Timing of Dataway command operations

During a command operation, command and data signals may take up either the "1" state or the "0" state. For convenience, Figure 9 shows only signals that take up the "1" state, but similar timing requirements apply to those that take up the "0" state.

The busy signal and the various command signals need not occur in exact synchronism, provided each is individually within the shaded areas of the diagram. Similar variation is permitted between the signals on the various data and status lines.

The W, R, Q and X signals are shown as being maintained until the end of the operation, but a broken line indicates the earliest time at which they are permitted to change as a result of actions initiated by strobe S2. During some operations, the Q signal may change at any time.

The L signal is shown for the particular case of a module that inhibits its L signal output in response to a command that does not clear the LAM source (see Sub-clause 5.4.1.3). The signal $L_1 = 1$ is therefore removed but reappears at the end of the operation.

Time-markers t_0 to t_{12} in Figure 9 indicate key points at which signal transitions are initiated or reach one of the threshold levels (0.8 V or 2.0 V).

At t_0 the transition of the busy signal to the "1" state is initiated. Command signals on the N, A and F lines also take up the "1" or "0" states as appropriate to the command.

At t_1 the busy signal has reached the 0.8 V threshold and all the command signals have reached the appropriate thresholds.

During the period $t_1 - t_3$ the addressed module responds to the command, and by t_2 the appropriate X, Q and data signals are initiated. By t_3 at the latest these signals have all reached the appropriate thresholds. Any L signals that are inhibited during the operation have reached the 2.0 V threshold by t_3 .

The transition of the S1 signal to the "1" state is initiated at t_3 and has reached the 0.8 V threshold by t_4 .

At t_5 the transition of the S1 signal to the "0" state is initiated and reaches the 2.0 V threshold by t_6 .

The transition of the S2 signal to the "1" state is initiated at t_6 and has reached the 0.8 V threshold by t_7 . Modules respond to S2 by changing the state of the R, Q and X signals.

The transition of the S2 signal to the "0" state is initiated at t_8 and has reached the 2.0 V threshold by t_9 , which is the end of the Dataway operation.

At t_9 the transition of the B signal to the "0" state is initiated and the command signals may also change from their established states.

At t_{10} the B signal and command signals have reached the 2.0 V threshold. During the period $t_{10} - t_{11}$ the module responds to the removal of the command. By t_{11} the transitions of the W, R, Q and X signals to the "0" state are initiated, and the inhibit is removed from the L signal. By t_{12} the L signal has reached the 0.8 V threshold and all other signals have reached the 2.0 V threshold.

Controllers shall initiate the transitions of the command and strobe signals at intervals not less than the minimum times shown in Figure 9. Modules shall respond to the command within the time $(t_1 - t_2)$ and to the strobes in the times $(t_4 - t_5)$ and $(t_7 - t_8)$. The electrical characteristics of the Dataway and connections from it into plug-in units must allow transitions between the two threshold levels to take place within the times $(t_0 - t_1)$, $(t_2 - t_3)$, etc.

The next Dataway operation shall not start before t_9 .

In the extreme case when the next operation starts at t_9 , the time-markers t_0 , t_1 , t_2 of the new operation coincide with t_9 , t_{10} , t_{11} of the previous operation. The command and data signals of one operation may thus be removed while those of the next operation are being established. The busy signal may be maintained continuously during a sequence of consecutive Dataway operations. Under suitable conditions, any command or data signals which have

the same state during successive operations may also be maintained. In the extreme case of successive operations with the same command and data, there could be a complete absence of signal transitions between t_0 and t_3 .

7.1.3.2 Timing of unaddressed Dataway operations

The sequence of events during an unaddressed clear or initialize operation is shown in Figure 10.

At t_0 the transition of the busy signal to the "1" state is initiated. In a clear operation, the transition of the C signal is also initiated at this time. In an initialize operation, the transitions of the Z and I signals are initiated.

By t_1 the B signal and, as appropriate, either Z and I or C have reached the 0.8 V threshold.

The interval $t_1 - t_6$ allows integration of the Z or C signals within the module if required.

At t_6 the transition of the S2 signal to the "1" state is initiated. The S2 signal is established and removed as described previously. (The S1 signal may be generated with timing relative to t_6 as shown in Figure 9.)

The S2 signal reaches the 2.0 V threshold by t_9 . The transitions of the B signal and C or Z to the "0" state are initiated at t_9 and reach the 2.0 V threshold by t_{10} . The inhibit signal may be removed at t_9 or, if possible, it is maintained in the "1" state as indicated by the broken line.

7.1.4 Digital signals on non-standard connections

Signals on the free bus-lines (contacts P1 and P2 at normal stations) shall be generated from intrinsic OR outputs and conform to the voltage levels of Table V. They shall conform to the current standards of either Table VII or Table VI, for read or write lines as appropriate (see also Sub-clause 5.6.1).

Signals on patch connections using contacts P3 to P5 at normal stations or P1 to P7 at the control station shall be generated from intrinsic OR outputs and shall conform to the voltage levels of Table V and the current standards of Table VII. Disconnected inputs shall take up the "0" state.

In Table VII, each input and output has an individual pull-up current source to compensate for leakage current in the "0" state. This allows flexibility in the number of inputs and outputs that can be patched together.

TABLE VII
Current standards for patch contacts

State of line	Current to and from patch connections	
	Outputs	Inputs
“1” state at +0.5 V	Units shall be capable of drawing more than 15 mA from connection when generating “1”	Unit shall not feed more than 2 mA into connection
	Unit shall not feed more than 300 μ A into connection when generating “0”	
“0” state at +3.5 V	Pull-up capability (current fed into connection) 100 μ A minimum 300 μ A maximum	

7.2 Other digital signals

The standards defined below should normally be used for all terminated and unterminated digital signals via connectors on the front panel and at the back of plug-in units above the Dataway. There may, however, be special circumstances requiring the use of other signals, for example, to suit a specific equipment with which the plug-in unit is closely associated.

7.2.1 *Unterminated signals*

Unterminated signals should conform to the standards set out in Table VIII, unless there are special reasons for using other standards.

Individual outputs shall be able to withstand, without damage, a short-circuit to ground. Outputs through multiway connectors need not withstand a short-circuit on all pins simultaneously.

Disconnected inputs shall take up the "0" state.

TABLE VIII
Unterminated signals

Outputs *	V_{out}	Logic "1"	Unit shall generate 0 V to +0.5 V
		Logic "0"	Unit shall generate +2.4 V to +5.5 V
	I_{out}	Logic "1" at +0.5 V	Unit shall draw > 16 mA from connection
		Logic "0" at +2.4 V	Unit shall feed > 6 mA into connection
Inputs	V_{in}	Logic "1"	Unit shall accept 0 V to +0.8 V
		Logic "0"	Unit shall accept +2.0 V to +5.5 V
	I_{in}	Logic "1" at +0.5 V	Unit shall feed < 2.0 mA into connection
		Logic "0" at +2.4 V	Unit shall feed current into connection or draw < 100 μ A from connection

* Not necessarily intrinsic OR.

7.2.2 *Terminated signals*

The characteristic impedance for terminated signals is 50 Ω . Signals terminated in 50 Ω should conform to the standards set out in Table IX, unless there are special reasons for using other standards.

Negative signs indicate currents flowing into an output circuit.

TABLE IX
Terminated signals

	Logic "0"	Logic "1"
Outputs shall deliver into 50 Ω :	–2 mA to +2 mA Preferred: –1 mA to +1 mA	–14 mA to –18 mA
Inputs shall accept:	–4 mA to +20 mA	–12 mA to –36 mA

7.3 *Analogue signals*

These signals, largely used for interconnections between units, should be standardized in the future.

8. Power line standards

The Dataway includes bus-lines for mandatory, additional and reserved power supplies.

Designers of plug-in units may assume that the mandatory lines (+24 V, +6 V, -6 V, -24 V and 0 V power return) are powered in every installation.

The additional bus-lines are provided for special requirements; for example, for accommodating NIM modules in a CAMAC crate.

Power is only supplied to these lines if expressly requested.

The voltages available to plug-in units at the contacts of each Dataway connector shall be within the tolerances specified in Table X. Individual plug-in units, and assemblies of plug-in units within a crate, shall not exceed the maximum current loadings specified in Table X.

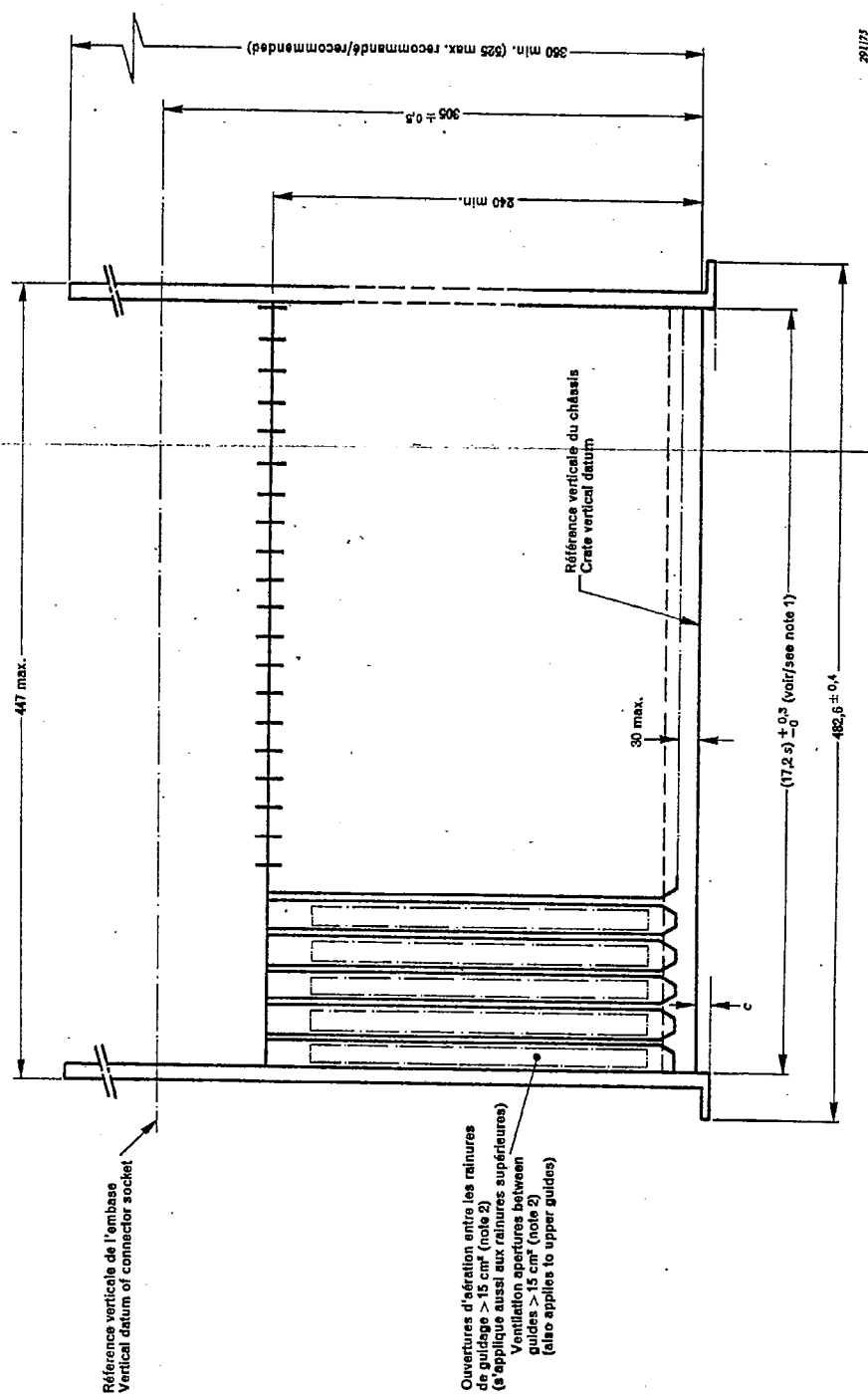
The Dataway power lines and any wiring from them to the point at which power supplies enter the crate shall be capable of carrying the maximum current loadings permitted in the crate. The resistance between any point on the Dataway 0 V power return bus-line and the point at which power supplies enter the crate shall not exceed 2 mΩ.

9. Environmental conditions of use

The system is intended for use in environments typically associated with laboratory instrumentation. The ambient temperature range has been provisionally defined as from 10 °C to 45 °C.

TABLE X
Power line standards

Nominal voltage on power line in crate	Voltage tolerance at Dataway connectors	Maximum current loads		Notes
		In the plug-in (per unit width) See Notes 1 and 3	In the crate See Note 2	
Mandatory +24 V d.c. +6 V d.c. -6 V d.c. -24 V d.c. 0 V	±1.0% ±2.5% ±2.5% ±1.0%	1 A 2 A 2 A 1 A	6 A 25 A 25 A 6 A	1. The current carried by each contact of the Dataway connector shall not exceed 3 A. 2. The total power dissipation in a crate without forced ventilation shall not exceed 200 W. 3. The power dissipation in each station shall not exceed 8 W in general or 25 W under special circumstances.
Additional (as required) +200 V d.c. +12 V d.c. -12 V d.c. 117 V a.c.	+60V, -20 V ±1.0% ±1.0%		0.1 A 0.5 A	Frequency 47 Hz to 63 Hz, to be obtained from an isolating transformer.



- Notes 1. — s = nombre de stations pour les tiroirs.
 2. — Ouvertures d'aération aussi longues que possible.
 3. — Généralement $c = 3,5 \pm 0,5$.
- Notes 1. — s = number of stations for plug-in units.
 2. — Ventilation apertures between guides to be as long as possible.
 3. — Typically $c = 3,5 \pm 0,5$.

Fig. 2. — Châssis — rainures de guidage inférieures — vue de dessus.
 Plan view of lower guides in crate.

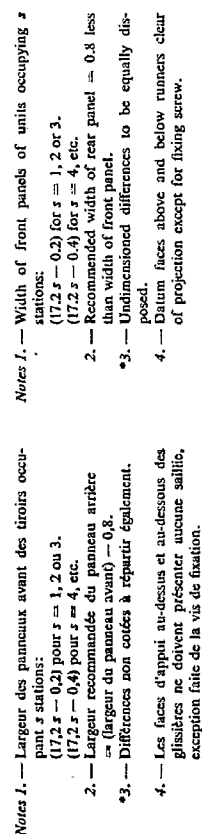
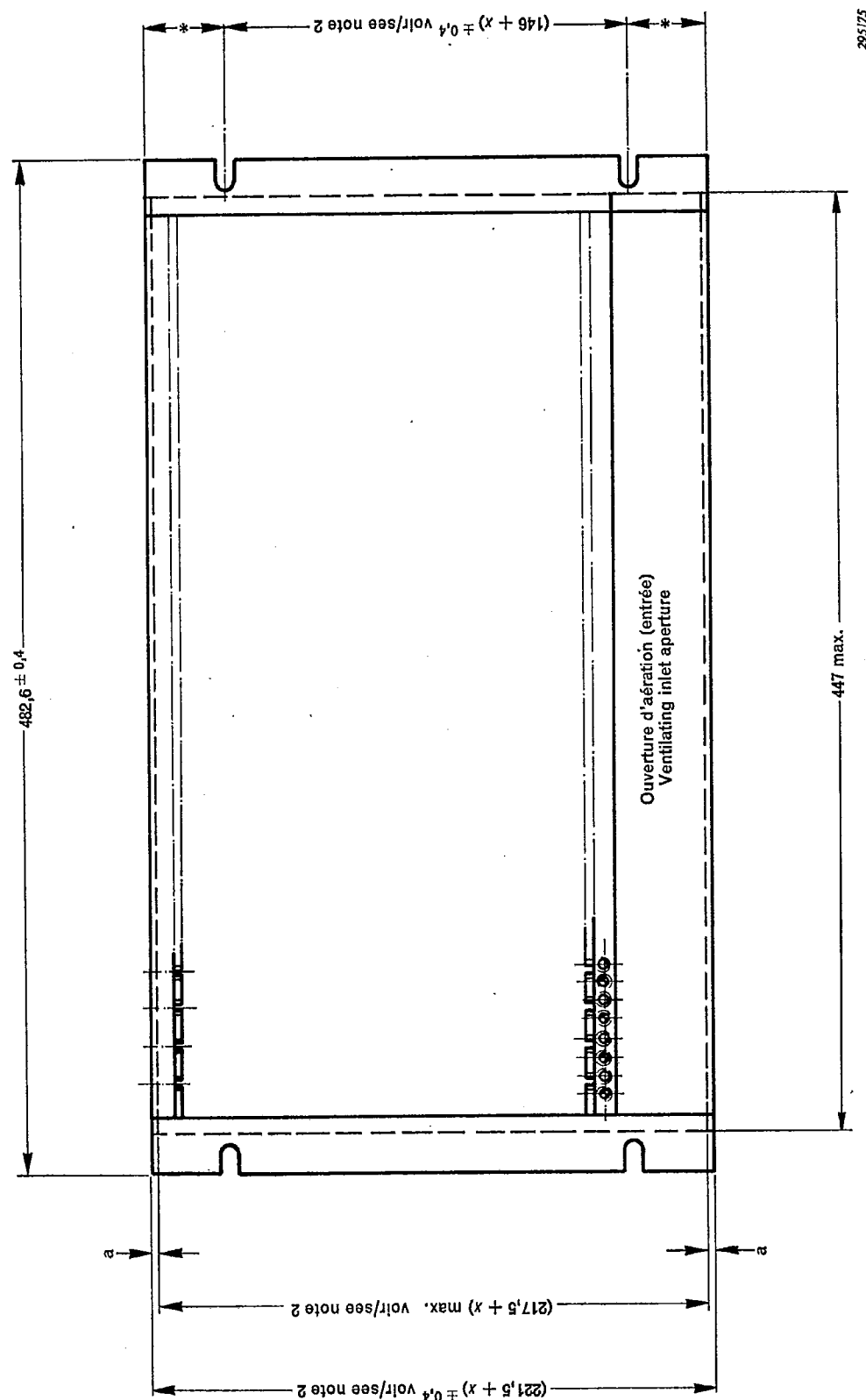


FIG. 4. — Tiroirs — vue de profil et vue arrière.
Plug-in unit, side and rear views.



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- Notes 1. — For all details not shown see Figure 1.
 2. — $x = 44,45 L$ where $L = 1, 2, 3$ etc.
 *3. — Undimensioned differences to be equally disposed.

FIG. 6. — Châssis — vue de la face avant avec 1U de ventilation.
 Ventilated crate — front view.

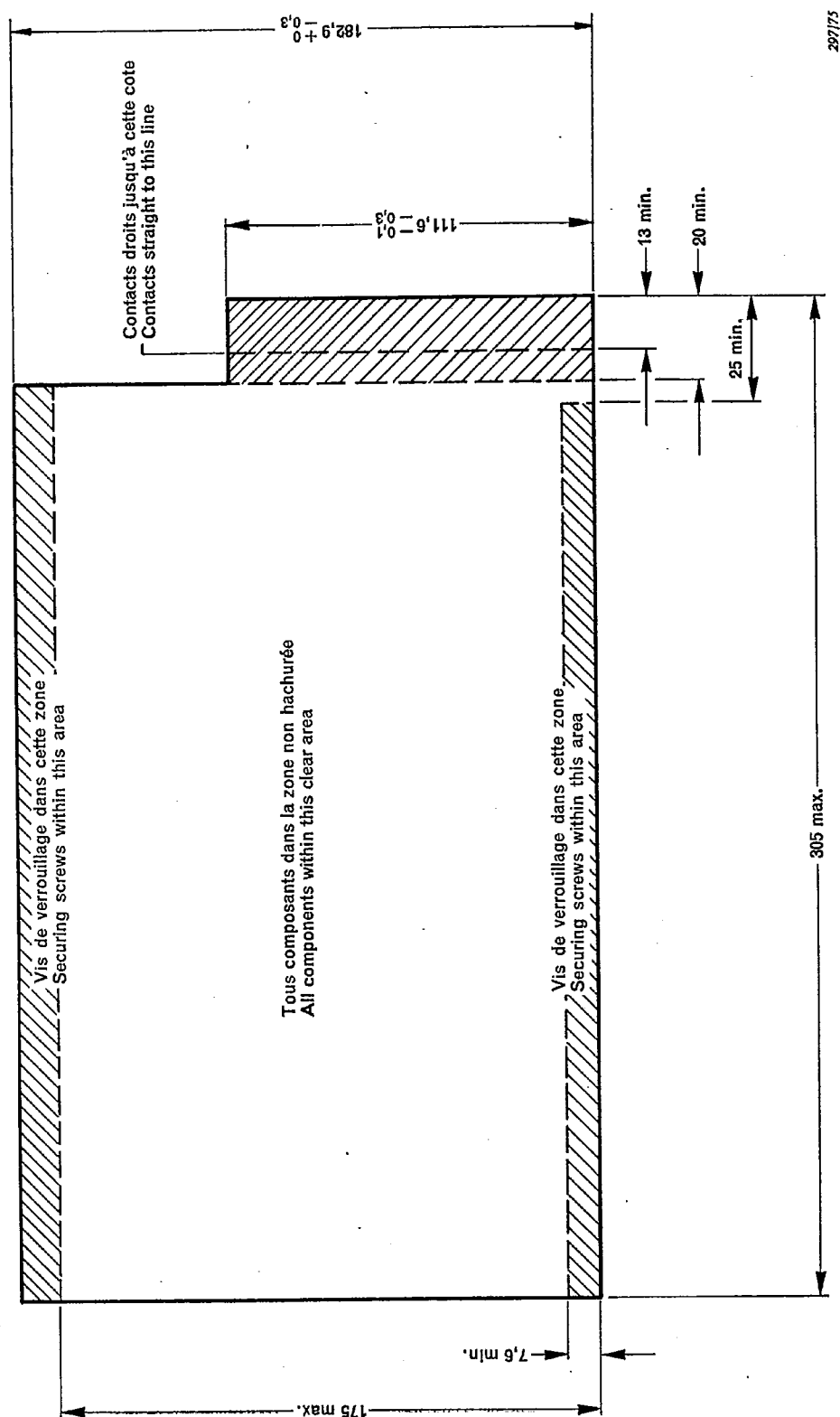
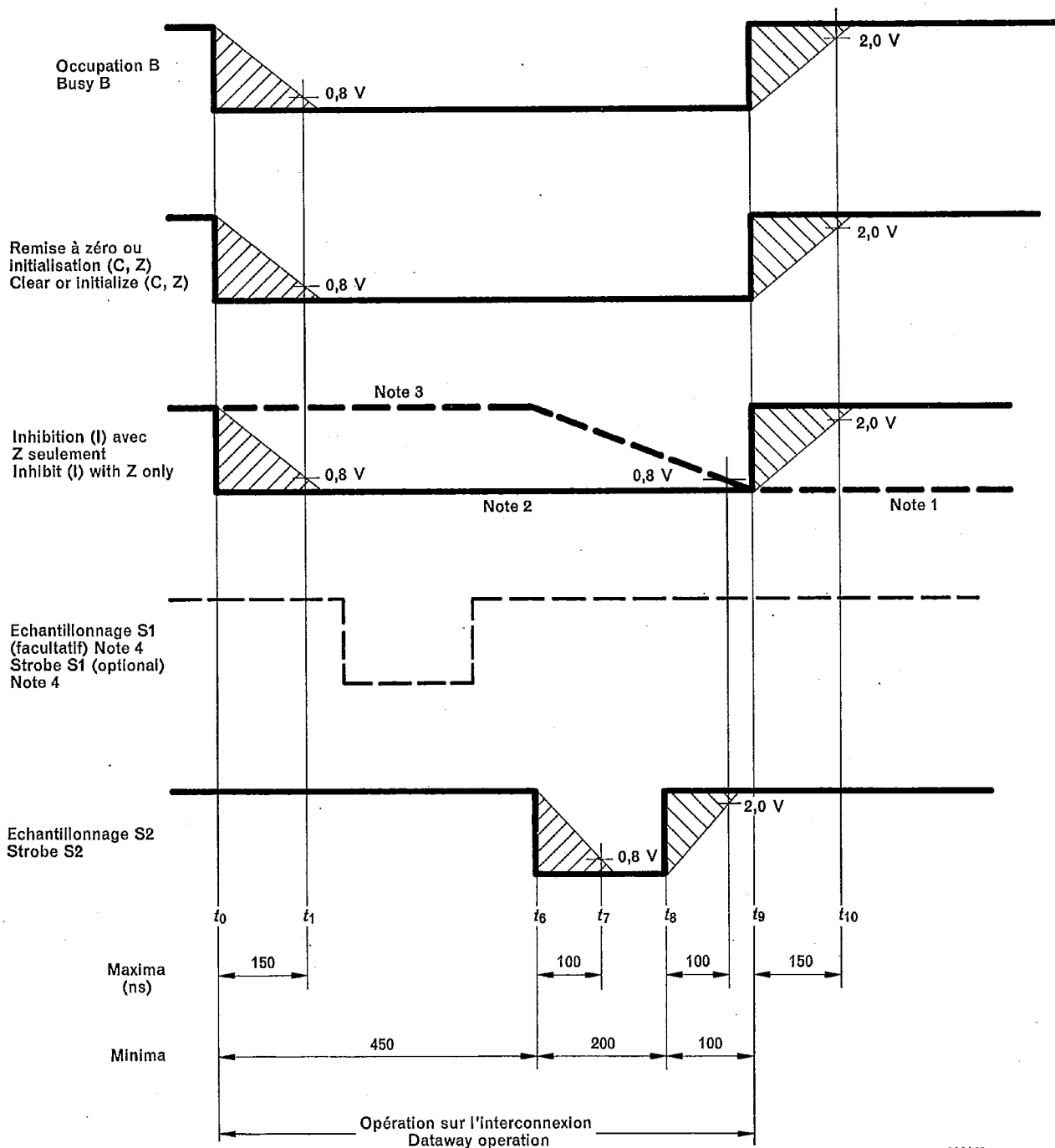


FIG. 8. — Carte type pour circuit imprimé.
Typical printed wiring card.



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Notes 1. — De préférence I est maintenu.
 2. — I accompagne Z.
 3. — I émis en réponse à Z.S2.
 4. — Autres temps comme dans la figure 9.

Notes 1. — I preferable maintained.
 2. — I accompanying Z.
 3. — I generated in response to Z.S2.
 4. — Other times as Figure 9.

FIG. 10. — Chronogramme d'une opération non adressée sur l'Interconnexion.
 Timing of a Dataway unaddressed operation.